

Loyalty Rewards — CRUD API Test Harness

Runs the full CRUD matrix against /odata/v4/loyalty (or) /apitestharness/.../odata/v4/loyalty · results are compared against the expected contract (docs/crud-test-results.md)

Locally (npm run watch) log in as the mock user you want to act as: **carol / pass** admin, **bob / pass** staff, **alice / pass** customer.
On the deployed CF app just open this page with your BTP account — the role flags of your logged-in user are shown in the banner above.
Read the matrix like this: run as **admin** (carol, or the BTP admin) → the main matrix passes; the **Cross-role** section only passes when a *customer-only* user runs it (as admin the ownership guard rightly lets the request through, so those four rows fail by design). Everything this page creates is throwaway test data.

signed-in user: **mrjayvasan@gmail.com** · admin=true staff=true customer=true · pointValueInr=0.5

Run all cases — 34 passed, 2 failed

Customers

Op	Case	Request	Expected	Status	Verdict
CREATE	staff onboards a new customer	POST /Customers {"name":"Harnes s msutgobi","email":"harness.ms utgobi@example.com"}	201 — staff may onboard anyone	201	PASS
CREATE	validation: missing name/email	POST /Customers {"name":"","ema il":"x"}	400 — mandatory fields	400	PASS
READ	list customers (admin/staff)	GET /Customers?\$top=5&\$select=c ustomerID,name,email,totalPoint s,tier	200 with array value	200	PASS
READ	read single by key	GET /Customers(customerID=c1a94 c08-bdf2-4b85-b9ac-9fcde003fa 84)	200 — key read	200	PASS
UPDATE	PATCH customer (no role has UPDATE — balances are system-owned)	PATCH /Customers(customerID=c1a 94c08-bdf2-4b85-b9ac-9fcde003fa 84) {"name":"Hacked msutgobi"}	403 — UPDATE not granted to any role	403	PASS
DELETE	DELETE customer	DELETE /Customers(customerID=c1 a94c08-bdf2-4b85-b9ac-9fcde003f a84)	403 — DELETE not granted to any role	403	PASS

Transactions

Op	Case	Request	Expected	Status	Verdict
CREATE	record Online purchase (points	POST /Transactions {"customerID _customerID":"c1a94c08-bdf2-4b8	201 — amount 1000, pointsEarned	201	PASS

	computed)	5-b9ac-9fcde003fa84", "channel": "Online", "price": 1000}	50 (rate 0.05)		
CREATE	record Store purchase with 20 points applied	POST /Transactions {"customerID": "c1a94c08-bdf2-4b85-b9ac-9fcde003fa84", "channel": "Store", "price": 500, "pointsApplied": 20}	201 — amount 490 (500 – 20×0.5), pointsEarned 14	201	PASS
CREATE	validation: invalid channel	POST /Transactions {"customerID": "c1a94c08-bdf2-4b85-b9ac-9fcde003fa84", "channel": "Phone", "price": 100}	400 — channel must be Online Store	400	PASS
CREATE	validation: unknown customer	POST /Transactions {"customerID": "00000000-0000-0000-0000-000000000000", "channel": "Online", "price": 100}	400 — no customer found	400	PASS
CREATE	validation: overspending applied points	POST /Transactions {"customerID": "c1a94c08-bdf2-4b85-b9ac-9fcde003fa84", "channel": "Online", "price": 100, "pointsApplied": 9999}	400 — insufficient points	400	PASS
READ	customer history (filter by customer)	GET /Transactions?\$filter=customerID eq c1a94c08-bdf2-4b85-b9ac-9fcde003fa84&\$select=txnDate,channel,amount,pointsApplied,pointsEarned	200 — 2 rows created above	200	PASS
READ	aggregate: points by channel	GET /Transactions?\$apply=groupby((channel),aggregate(pointsEarned with sum as total))	200 — per-channel totals (Online and Store rows, numeric sums)	200	PASS
UPDATE	PATCH transaction (immutable ledger)	PATCH /Transactions(txnID=29ef64f6-831c-44f3-bc34-a3578f96189a) {"amount": 1}	403 — UPDATE not granted (audit integrity)	403	PASS
DELETE	DELETE transaction	DELETE /Transactions(txnID=29ef64f6-831c-44f3-bc34-a3578f96189a)	403 — DELETE not granted (audit integrity)	403	PASS

Redemptions

Op	Case	Request	Expected	Status	Verdict
CREATE	redeem 10 points	POST /Redemptions {"customerID": "c1a94c08-bdf2-4b85-b9ac-9fcde003fa84", "pointsUsed": 10, "remarks": "harness voucher"}	201 — balance reduced by 10	201	PASS

CREATE	validation: redeem more than balance	POST /Redemptions {"customerID_customerID":"c1a94c08-bdf2-4b85-b9ac-9fcde003fa84","pointsUsed":99999}	400 — insufficient points / never negative	400	PASS
CREATE	validation: non-positive pointsUsed	POST /Redemptions {"customerID_customerID":"c1a94c08-bdf2-4b85-b9ac-9fcde003fa84","pointsUsed":0}	400 — must be a positive integer	400	PASS
READ	redemption history for customer	GET /Redemptions?\$filter=customerID eq c1a94c08-bdf2-4b85-b9ac-9fcde003fa84&\$select=redeemDate,pointsUsed,remarks	200 — 2 rows (part-payment + standalone)	200	PASS
UPDATE	PATCH redemption	PATCH /Redemptions(redeemID=05beaaa8-a0c6-4202-8f39-a1f2bbfc24f1) {"pointsUsed":1}	403 — UPDATE not granted	403	PASS
DELETE	DELETE redemption	DELETE /Redemptions(redeemID=05beaaa8-a0c6-4202-8f39-a1f2bbfc24f1)	403 — DELETE not granted	403	PASS

RewardPolicies

Op	Case	Request	Expected	Status	Verdict
READ	list policies	GET /RewardPolicies	200 — Online 0.05, Store 0.03 seeded	200	PASS
CREATE	duplicate channel policy rejected (unique)	POST /RewardPolicies {"channel":"Online","pointsPerCurrencyUnit":0.9}	409 — one policy per channel	409	PASS
CREATE	validation: invalid channel	POST /RewardPolicies {"channel":"Catalog","pointsPerCurrencyUnit":0.1}	400 — channel must be Online Store	400	PASS
UPDATE	admin adjusts Store rate (keyed by policyID)	PATCH /RewardPolicies(policyID=9f8f6f1a-2222-4000-8000-000000000002) {"pointsPerCurrencyUnit":0.04}	200 — rate updated, cache reloaded	200	PASS

TierThresholds

Op	Case	Request	Expected	Status	Verdict
READ	list thresholds	GET /TierThresholds	200 — Bronze/Silver/Gold seeded	200	PASS
CREATE	add harness tier above the test customer	POST /TierThresholds {"tier":"Harness","minLifetimePoints":10}	201 — and existing customer tiers recomputed (lifetime ≥ 10)	201	PASS

READ	verify recompute: customer now tier Harness	GET /Customers(customerID=c1a94c08-bdf2-4b85-b9ac-9fcde003fa84)?\$select=tier	200 — tier === Harness (recomputed without a purchase)	200	PASS
UPDATE	update threshold minimum	PATCH /TierThresholds(tier='Harness') {"minLifetimePoints":500000}	200 — minimum updated	200	PASS
READ	verify recompute back to Bronze	GET /Customers(customerID=c1a94c08-bdf2-4b85-b9ac-9fcde003fa84)?\$select=tier	200 — tier back to Bronze	200	PASS
DELETE	remove harness tier	DELETE /TierThresholds(tier='Harness')	204 — cleanup	204	PASS

Cross-role (as alice)

Op	Case	Request	Expected	Status	Verdict
CREATE	customer buys for a FOREIGN account	POST /Transactions {"customerID_customerID": "c1a94c08-bdf2-4b85-b9ac-9fcde003fa84", "channel": "Online", "price": 100}	403 — ownership guard precedes validation	201	FAIL
CREATE	customer buys for OWN seeded account	POST /Transactions {"customerID_customerID": "b1a7e6d0-1111-4000-8000-000000000001", "channel": "Online", "price": 100}	201 — self-service purchase allowed	201	PASS
CREATE	customer redeems a FOREIGN account	POST /Redemptions {"customerID_customerID": "c1a94c08-bdf2-4b85-b9ac-9fcde003fa84", "pointsUsed": 1}	403 — ownership guard precedes validation	201	FAIL
CREATE	customer re-registers own email	POST /Customers {"name": "Alice Dup", "email": "alice@example.com"}	409 — one account per email	409	PASS

getUserInfo

Op	Case	Request	Expected	Status	Verdict
FUNCTION	role flags + server-owned point value	GET /getUserInfo()	200 — isAdmin/isStaff/isCustomer flags and pointValueInr 0.5	200	PASS